

Srihari Srinivasan

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EDUCATION

University of Warwick

BEng Computer Systems Engineering

- **First Class** in Year 1, including **90%** in Engineering Mathematics. **Warwick Undergraduate Global Excellence Scholarship** (50% tuition, academic merit) and **Warwick Award (Silver)**.
- Year 2 modules include Artificial Intelligence, Signal Processing and Advanced Computer Architecture.

Coventry, United Kingdom

Sept 2025 – Jul 2028

GEMS New Millennium School

A-Level A*AAB (A* Computer Science); AS-Level AAAAA incl. Mathematics and Further Mathematics

Dubai, UAE

Aug 2021 – Jun 2025

RESEARCH INTERESTS

Interpretability of deployed models; data poisoning and backdoors in LLMs; the ML systems underneath (GPU kernels, compilers).

PUBLICATIONS

- **S. Srinivasan**. “Depth-Blind by Design: An Interpretability Autopsy of a Frozen Kepler Transit Classifier on TESS.” *NeurIPS 2026 Workshop on Interpretability for Discovery (Interp4Discovery)*. Accepted, sole author.
- **S. Srinivasan**. “Understanding User Perception of Biometric Privacy in the Era of Generative AI.” *4th Int. Conf. on Communication, Computing and Industry 6.0 (C2I6)*, IEEE, 2023. Sole author; cited 6 times, 480 full-text views. doi:10.1109/C2I659362.2023.10430931

RESEARCH EXPERIENCE

Interpretability of an exoplanet classifier | *NeurIPS 2026 workshop* | github.com/SrihariSr/Stellaris

May – Aug 2026

- Froze a 9.71M-parameter two-branch CNN trained on NASA Kepler labels (0.945 held-out PR-AUC, against 0.918 for an XGBoost baseline on the same views) and deployed it unchanged on 5,578 TESS targets, then asked where its signal lives, which stage loses planets and what it cannot see.
- Over ten seeds, a 1.32M-parameter local-view network matched the full 9.71M one (0.943 against 0.944 PR-AUC, Welch $p = 0.83$), bounding what the 8.4M-parameter global branch adds at 0.005 PR-AUC with 95% confidence. Of 417 missed confirmed planets, the network alone lost more than the rule-based vetting alone at the deployed thresholds (179 against 131, McNemar $p = 0.007$).
- On 474 faint stars, 41.6% of search periods fell within 10% of TESS’s 13.7-day orbital period, a 7.6-fold excess over chance, including all four detections the network scored above 0.9; its median score was 0.030 on these against 0.032 elsewhere. Identified that the input normalisation discards absolute transit depth, the leading candidate explanation. A released script regenerates every reported number.

Detecting training-data poisoning in LLMs | *Supervisor: Prof. Ligang He, University of Warwick*

Aug 2026 – present

- In preparation for ICML 2027: a method for detecting backdoors planted in pretraining data, to be tested by inference on public checkpoints of Gaperon, an open model suite whose pretraining data was deliberately seeded with harmless trigger-based poisoning.

ML SYSTEMS & OPEN SOURCE

PyTorch | Faster native Metal matmul kernels for the MPS backend | pytorch/pytorch#198182

Sept 2026 – present

- Measured PyTorch’s native Metal matmul kernels at 6.0x to 9.8x slower than Apple’s MPSGraph on typical shapes on an M4 Max (median 7.9x) and proposed a rewrite; a maintainer asked for it as a gist, to be weighed against Apple’s MPP path.
- Wrote two kernels chosen by data type, register-blocked for float32 and `simdgroup_matrix` for float16 and bfloat16, with a split-K path for small outputs: 2.1x to 2.9x faster in float32 and 2.4x to 4.3x in float16 and bfloat16 on square shapes from 1024 upwards, and about 7x on 25x40000x25, where they beat MPSGraph by 1.2x and MPP by 5.4x to 6.3x. All 48 correctness cases pass against a float64 reference. Losses reported too: 0.73x to 0.87x at 256x256x256, and still about 2x behind MPSGraph on large square shapes.

Forge | ML library, compiler and GPU kernels from scratch | pip install forge-dl | github.com/SrihariSr/Forge

Mar 2026 – present

- Reverse-mode automatic differentiation with every operation checked against finite differences, used to train a 551K-parameter GPT data-parallel across 8 CPU cores, and a compiler that plans operator fusion (2.36x on 16M-element chains), generates C, compiles it with GCC and loads it through ctypes. Published on PyPI (v0.3.0).
- Took a 768x768 CPU matrix multiply from 348ms to 3.8ms (91x) in portable C with pthreads, within 1.5x of Apple’s Accelerate BLAS: loop reordering gave 12x and threading 8.4x, while hand-written NEON intrinsics measured slower than Clang’s auto-vectorised loop (48ms against 32ms) and were dropped.
- Three CUDA matmul kernels in C++ (naive, shared-memory tiled, register-blocked): the last reaches 2,718 GFLOPS on a Tesla T4, within 1.9x of cuBLAS at every size tested, each verified against a double-precision reference on six shapes.

EXPERIENCE

AI/ML Intern

Tech Mahindra

Bengaluru, India

Jul – Aug 2026

- Took over and maintained a large existing codebase from an outgoing engineer: a graph neural network for root cause analysis that models telecommunication network elements and their dependencies to find where a cascading failure began, at 99.46% top-1 and 99.88% top-3 accuracy on the originating node (AUROC 0.9322).

PERSONAL PROJECTS

Aether | Transformer language model trained from scratch | github.com/SrihariSr/Aether

Mar – Apr 2026

- A 63.5M-parameter decoder-only transformer in PyTorch reaching validation perplexity 63 on OpenWebText, with BPE tokenisation parallelised across 12 CPU cores into memory-mapped files, sustaining about 11,150 tokens/sec on one laptop.

Foundry | Offline classifiers from one sentence | *Push to Prod, Anthropic & Elevation Capital* | github.com/SrihariSr/Foundry

Aug 2026

- Turns a one-sentence task description into a self-contained 691KB classifier trained on data Claude generates. On a held-out disaster-triage set it scores 0.95 accuracy against 0.97 for Claude Haiku 4.5, at 2.4ms against 842ms per decision.

AWARDS & PATENTS

- **1st place**, Warwick Entrepreneurs Hackathon, Feb 2026.
- **Patent filed**, Indian Patent Office, Sept 2023: a smart polling machine for election integrity and efficiency.

TECHNICAL SKILLS

Languages: Python, C, C++, CUDA, Metal Shading Language, Java, SQL, Bash, MATLAB

ML & Research: PyTorch, NumPy, SciPy, scikit-learn, XGBoost, pandas; multi-seed ablations, significance testing, reproducible pipelines

Systems: cuBLAS, Apple Metal/MPS and Accelerate, pthreads, GCC, Clang, Linux, Git